

Obstetric Fistula (VVF/RVF)

Tutorial

Week 5 · Bleeding, cancer & fertility

Gynaecology

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An obstetric fistula is an abnormal opening between the genital tract and the bladder (VVF) or rectum (RVF). It is usually caused by ischaemic pressure necrosis from prolonged obstructed labour, and increasingly by surgical injury at CS or hysterectomy. It presents as continuous leakage of urine or faeces per vaginam. It is preventable by timely delivery and immediate catheterisation, and curable in most women by skilled, tension-free vaginal repair followed by uninterrupted bladder drainage.

Must know before the ward round

1. Commonest cause in Uganda is prolonged obstructed labour, where the impacted head crushes the bladder base and vagina against the pubis. The necrotic slough separates at about 3–10 days postpartum, so leakage starts days after delivery. Iatrogenic (CS/hysterectomy) fistulae now make up nearly a third of repairs.
2. Immediate leakage after surgery suggests a direct surgical injury. Leakage that starts days after delivery suggests an ischaemic (obstructed labour) fistula.
3. Continuous leakage with normal voiding suggests a ureterovaginal fistula. Cyclical haematuria with amenorrhoea and continence suggests a vesicouterine fistula (Youssef syndrome).
4. Dye test: instil dilute methylene blue into the bladder with swabs in the vagina. Blue = VVF. Wet but not blue = ureterovaginal (confirm with the three-swab or double-dye test).
5. A fresh fistula needs a Foley catheter on continuous free drainage for up to 4–6 weeks with high fluid intake. Small fresh fistulae may close without surgery.
6. Repair principles are good exposure, wide mobilisation, tension-free watertight closure in separate layers and uninterrupted bladder drainage afterwards (WHO 2018: 7–10 days for a simple fistula).
7. Goh classification uses the distance of the distal edge from the external urinary meatus (type 1 more than 3.5 cm to type 4 less than 1.5 cm), size (a–c) and scarring (i–iii). Type 4 (urethral involvement) has the worst continence outcome.
8. After repair: no intercourse for about 3 months, family planning, and deliver every future baby by elective CS in hospital.
9. Rule out the mimics first: palpate the bladder and check a post-void residual (over 150 mL = retention with overflow), and do a dye test before labelling any leakage a fistula.
10. Three-swab test: upper swab wet but not blue = ureterovaginal; middle swab blue = mid-vaginal VVF; upper swab blue = juxtacervical or vault; lowest swab only = urethral (or dye spill).
11. Always document the distance from the external urethral meatus, the size, the scarring, the urethral length and a rectal examination, so the fistula can be Goh-graded and triaged as simple or complex.

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In a nutshell

- **Obstetric fistula** is a hole between the vagina and the **bladder or urethra** (vesicovaginal fistula, **VVF**) or the **rectum** (rectovaginal fistula, **RVF**). It is usually caused by **days of obstructed labour** and, increasingly, by **surgical injury at caesarean section or hysterectomy**.
- The result is **continuous, uncontrollable leakage of urine and/or faeces**. That leads to wet, smelly clothes, **ammoniacal dermatitis**, and then **divorce, stigma, isolation, poverty and depression**. The baby has usually died in the labour that caused the fistula.
- Uganda still has many thousands of women living with fistula, although national survey data suggest the prevalence is falling. **Each fistula is a failure of the health system**: a delay in deciding to seek care, in reaching care, or in receiving a CS.
- The **one thing never to forget**: after any prolonged or obstructed labour, **put in a catheter and leave it for 7–14 days**. For a fresh fistula, **continuous bladder drainage for up to 4–6 weeks** can close a small one without surgery.
- Most fistulae can be **repaired vaginally** by a trained surgeon, with closure rates of about 80–90%. Residual **stress incontinence** remains a common problem, especially when the urethra is involved.

Definitions & classification

By anatomy

Type	Communication	Typical clue
Urethrovaginal	Urethra to vagina	Leakage may be positional; the closing mechanism is involved, so there is a high risk of post-repair stress incontinence
Vesicovaginal (VVF)	Bladder to vagina (mid-vaginal, juxtacervical, vault)	Continuous leakage day and night ; little or no voiding
Vesicocervical / vesicouterine	Bladder to cervix or uterus, usually after CS	Youssef syndrome : cyclical haematuria (menouria), amenorrhoea, and often continence
Ureterovaginal	Ureter to vagina, usually after CS or hysterectomy	Continuous leakage plus normal voiding (the other ureter still fills the bladder). May have loin pain or hydronephrosis
Rectovaginal (RVF)	Rectum to vagina	Flatus or faeces per vaginam. It may follow obstructed labour, third- or fourth-degree tears (low RVF), or sexual violence
Combined VVF + RVF	Both	Severe obstructed-labour injury complex

By cause

Cause	Mechanism	When leakage starts
Ischaemic (obstetric)	Prolonged obstructed labour, where the head compresses soft tissue against the bone, causing pressure necrosis	3–10 days postpartum , when the slough separates
Iatrogenic (surgical)	Bladder injury at CS (especially repeat, emergency or for obstructed labour), caesarean hysterectomy , abdominal or vaginal hysterectomy, destructive operations , forceps, symphysiotomy	Often immediate or within 1–2 days of surgery; ureteric injuries after 1–2 weeks

Cause	Mechanism	When leakage starts
Traumatic	Sexual violence, penetrating injury, harmful traditional practices (e.g. gishiri cuts), foreign bodies	Variable
Other	Cervical cancer, radiation , schistosomiasis, LGV, TB (rare)	Gradual

Waldijk classification (surgical)

- **Type I:** does **not** involve the urethral closing mechanism.
- **Type II: involves the closing mechanism.**
 - **A:** without (sub)total urethral involvement. **B:** with (sub)total urethral involvement.
 - **a:** without a circumferential defect. **b:** with a circumferential defect.
- **Type III:** ureteric and other exceptional fistulae.

Surgical difficulty rises, and closure and continence rates fall, from **type I through type IIBb**.

Goh classification (predicts outcome better)

Component	Grade	Definition
Site (distal edge from the external urinary meatus, EUM)	Type 1	More than 3.5 cm
	Type 2	2.5–3.5 cm
	Type 3	1.5 to under 2.5 cm
	Type 4	Under 1.5 cm (involves the urethra)
Size (largest diameter)	a	Under 1.5 cm
	b	1.5–3 cm
	c	Over 3 cm
Scarring	i	None or mild fibrosis; vaginal length over 6 cm, normal capacity
	ii	Moderate or severe fibrosis; reduced vaginal length or capacity
	iii	Special considerations: post-radiation, ureteric involvement, circumferential defect, previous repair

For example, a "**Goh 2bi**" fistula has its distal edge 3 cm from the EUM, is 2 cm across, and has minimal scarring. It has a good prognosis. A **type 4** fistula (with the urethra involved) is the most likely to fail or leave stress incontinence.

EXAM FAVOURITE

Simple vs complex fistula (WHO/FIGO working definition). A **simple** fistula is single, small (under about 3–4 cm), not involving the urethral closing mechanism, not circumferential, with minimal scarring, no ureteric involvement, and a first repair. It can be repaired by a trained generalist. A **complex** fistula (large, urethral, circumferential, scarred, previous failed repair, combined VVF/RVF or ureteric) needs a **fistula specialist centre**.

Epidemiology & Uganda context

- WHO estimates **50,000–100,000 new obstetric fistulae worldwide each year**. Recent estimates suggest about half a million women are living with fistula, down from about 2 million two decades ago, but still overwhelmingly in **sub-Saharan Africa and South Asia**.
- **Uganda:**
 - The **Uganda Demographic and Health Survey (UDHS)** shows the proportion of women reporting fistula symptoms falling from about 3% (2006) to about 1% (2016). The highest burden is in the **Tooro and Ankole** regions.
 - Ministry of Health estimates have ranged from **about 100,000 to 200,000 women living with fistula**, with **about 1,900 new cases a year**.
 - A multicentre Ugandan cohort of **510 repairs (2010–2021)** found **66% ischaemic** fistulae and nearly **one third iatrogenic or other**. Successful **closure with continence was about 81%**.
- **The rising iatrogenic share** reflects more CS being done, often late, on tired surgeons and oedematous tissue, by less experienced hands at district level. Every CS for obstructed labour is also a risk to the bladder.
- **Why women still get fistula:** child marriage and adolescent pregnancy, short stature and a small pelvis, home delivery, poverty, distance and transport problems, poor labour monitoring, delayed referral, and HC IIIs without theatres.

UGANDA CONTEXT

Uganda treats fistula through **regional fistula centres and outreach "fistula camps"**. About **29 centres across 24 districts** have been set up, including national and regional referral hospitals and mission hospitals.

- **Kitovu Hospital (Masaka)** runs camps several times a year.
- **TERREWODE** (Soroti) runs a dedicated **women's community fistula hospital** and a reintegration programme.
- The **Fistula Foundation Treatment Network** was launched in Uganda with the Ministry of Health and partners.

Surgery is usually **free** to the woman, funded by the MoH, UNFPA and NGOs. Your job on the ward is to **identify, catheterise, counsel and link her** to the nearest centre. Know when the next camp is and who the regional fistula surgeon is.

Pathophysiology

The obstructed labour injury

1. The fetal head is **impacted in the pelvis for hours to days**.
2. Soft tissues trapped between the head and the **pubic symphysis** (anteriorly) or the **sacrum** (posteriorly) lose their blood supply. This is **ischaemia**.
3. The **bladder base, trigone, bladder neck, urethra** and anterior vaginal wall die (pressure necrosis). Posteriorly, the **rectum** can necrose.
4. Over **3–10 days** the dead tissue sloughs, leaving a **hole surrounded by oedema and infection**. The edges then **scar (fibrose)** over weeks.
5. Where the head was impacted predicts where the fistula forms:
 - At the **inlet**: high, juxtacervical or vesicouterine injury.
 - At the **mid-pelvis**: mid-vaginal VVF, the classic type.

- At the **outlet**: urethral or bladder-neck injury, plus RVF.

The "obstructed labour injury complex" (Arrowsmith): a field injury, not just a hole

The same ischaemia damages much more than the bladder:

- **Urethral loss** (causes stress incontinence even after closure).
- **Vaginal scarring and stenosis** (painful sex, infertility).
- **Ureteric involvement**.
- **Foot drop** (lumbosacral trunk or common peroneal nerve injury).
- **Pubic bone osteitis**.
- **Secondary amenorrhoea** (stress, malnutrition, Sheehan's syndrome after PPH).
- **Chronic dermatitis**.
- **Bladder stones**.
- **Infertility**.
- **Stigma, depression and poverty**.

This explains why "the fistula is closed but she still leaks": the continence mechanism (urethra and bladder neck) was destroyed along with the bladder base.

Why catheter drainage can heal a fresh fistula

A **decompressed bladder** does not stretch the defect, and urine is diverted away from the wound. The raw, granulating edges can then **contract and epithelialise**. This works only while the fistula is **small and fresh**. Once the edges are fibrotic and epithelialised (after weeks), spontaneous closure is unlikely.

Why surgery follows the principles it does

- **Tension-free closure**: tension causes ischaemia, which breaks the suture line down.
- **Wide mobilisation** of the bladder from the vagina allows tension-free closure.
- **Separate, non-overlapping layers** (bladder, then vagina): reduce re-fistulation.
- **Well-vascularised tissue** (e.g. a **Martius fat-pad flap** in scarred cases): brings in blood supply.
- **Uninterrupted drainage**: a **blocked catheter** lets the bladder distend and blows the repair apart. That is why nurses check the catheter hourly.

Clinical acumen

Focused history

Question	Why it matters
How long was the labour? Where? How was she delivered? Was the baby alive?	Days of labour, a stillbirth and a CS or destructive delivery suggest an ischaemic fistula .
When did leakage start?	Immediately after surgery : direct surgical injury (bladder or ureter). 3–10 days after delivery : ischaemic slough. 1–3 weeks after CS or hysterectomy : consider a ureteric injury.
Is the leakage continuous, or only on coughing or standing?	Continuous suggests fistula. Only on stress suggests stress incontinence. Only on urgency suggests overactive bladder.
Does she still pass urine normally?	Normal voiding with constant leakage suggests a ureterovaginal fistula .
Faeces or flatus per vaginam?	RVF (ask directly, because women may be ashamed).

Question	Why it matters
Cyclical haematuria? Periods?	Vesicouterine fistula (Youssef syndrome). Amenorrhoea is also common after obstructed labour.
Leg weakness, dragging the foot	Foot drop, part of the injury complex.
Previous repairs, and where	Scarring, and the need for specialist referral.
Social situation: husband, family, livelihood, isolation	Guides counselling, reintegration and support for transport.
Contraception and pregnancy wishes	Timing of repair and future delivery planning.
Mood, suicidal thoughts	Depression is very common.
HIV status, nutrition, anaemia	Affect healing and fitness for surgery.

Examination

- **General:** nutrition, pallor, hydration, gait (**foot drop**), mood, and the smell of urine.
- **Vulva and thighs:** **ammoniacal dermatitis** (red, excoriated, sometimes with papules or hyperkeratosis), wetness, and pooling of urine.
- **Speculum:** use a **Sims speculum** in the **left lateral or lithotomy** position, with good light (a head torch).
 - Locate the fistula.
 - Estimate its **size** (in fingertips or cm) and measure the **distance from the EUM**.
 - Look for **scarring and vaginal length**.
 - Assess the cervix.
 - Look for **sloughs**.
- **Digital:** feel the defect and the rim, and gauge how mobile the tissues are. Pass a **metal catheter or sound** through the urethra: can it be felt in the vagina through the fistula? Is the urethra intact and how long is it?
- **Rectal examination:** look for an RVF, check sphincter integrity (look for a cloaca, or a missing perineal body), and assess tone.
- **Abdomen:** bladder, renal angle tenderness (ureteric obstruction), and a CS scar.

Red flags

RED FLAG

- **Fever, loin pain or a flank mass after hysterectomy or CS:** ureteric injury with obstruction or urinoma.
- **Sepsis or peritonitis** after a difficult CS: bladder or bowel injury with urine or faeces leaking into the peritoneum.
- **Anuria:** bilateral ureteric injury or AKI. This is an urgent urology problem.
- **Suicidal ideation**, or severe depression.
- **Necrotic, foul slough** with systemic illness: needs debridement and antibiotics.
- **A fistula with an ulcerated, friable, hard mass:** **cervical cancer** until proven otherwise. Biopsy it.

Thinking like a clinician

1. **"She leaks since delivery." Is it really a fistula?** Separate **continuous** leakage (fistula), **stress** leakage (weak pelvic floor or urethral damage), **urgency** leakage (overactive bladder, UTI) and **overflow** (retention after delivery, epidural, pain). **Overflow incontinence** after a prolonged labour is common and treatable with a catheter. Always palpate for a **distended bladder** and do a **dye test** before labelling a fistula.

2. **"The dye test is negative but she leaks continuously"**: think **ureterovaginal fistula**. Do the three-swab or double-dye test and scan the kidneys.
3. **"She is continent but has blood in her urine with her periods"**: a **vesicouterine** fistula, usually after a CS.
4. **"The fistula is fresh"**: this is an **opportunity, not a disappointment**. Catheterise now. Don't send her home to wait 3 months without a catheter.
5. **Fistula is a symptom of the whole system**: document the delays (the three-delay model) and report the case for MPDSR-style review.
6. **Common student mistakes**:
 - forgetting the **rectal examination**
 - not asking about **faecal leakage**
 - not measuring the **distance from the EUM**
 - ignoring **foot drop** and **mood**
 - calling any post-delivery leakage a "VVF" without a dye test

Clinical workup

Fistula is usually a **planned, dignified outpatient or ward workup**, not a resuscitation. The exceptions are the woman seen days after an obstructed labour or a difficult CS, who may be septic, dehydrated or have urine in her peritoneum. The workup answers five questions: **Is it really a fistula (or overflow, stress or urge leakage)? Which organ is leaking (bladder, urethra, ureter, uterus, rectum)? Where is it and how big and scarred is it (Goh)? Is it fresh enough for catheter drainage? Is she fit and supported for repair?**

Step 0: First 5 minutes

1. **Greet her privately and with dignity**. Offer a pad and a seat she will not worry about wetting. Many women have been sent away or mocked; the first minute decides whether she tells you the whole story.
2. **Vital signs**: temperature, pulse, BP, RR. **Fever, tachycardia or hypotension** in the first weeks after obstructed labour or CS: look for **puerperal sepsis, pelvic abscess, necrotic slough** or **urinary ascites** (bladder injury leaking into the peritoneum).
3. **Abdomen in 30 seconds**: distension, peritonism (urine or faeces leaking into the peritoneum), a **tender flank** or **loin mass** (obstructed ureter or urinoma after CS or hysterectomy), a **palpable bladder** (retention with overflow, not a fistula).
4. **Is she passing any urine at all? Anuria** after pelvic surgery = possible **bilateral ureteric injury** or AKI: urgent U&E, renal ultrasound and a urologist the same day.
5. **Put a Foley catheter in now** (16–18 F, free drainage, never clamped) if the fistula is fresh (within about 4–6 weeks of delivery), or if the bladder is distended. This is both a diagnostic and a therapeutic step.
6. **Screen mood and safety**: ask directly about hopelessness and thoughts of self-harm. Depression is common; suicidal thoughts are a red flag needing same-day support.

RED FLAG

Fever, peritonism, loin pain or anuria after a CS or hysterectomy is a **surgical emergency** (bladder, bowel or ureteric injury). This is not a "fistula for the next camp": call the senior surgeon and urologist.

Step 1: Focused history

Ask	Why it changes the plan
The index labour : how long, where, how delivered (SVD, CS, destructive), baby alive?	Days of labour + stillbirth = ischaemic fistula; CS or hysterectomy = consider iatrogenic (bladder or ureter)

Ask	Why it changes the plan
When did the leaking start?	Immediately after surgery = direct surgical injury; 3–10 days after delivery = ischaemic slough; 1–3 weeks after CS/hysterectomy = think ureter
Continuous, or only on coughing, or with urgency?	Continuous = fistula; effort only = stress incontinence; urgency = OAB or UTI
Does she still void normally?	Continuous leakage + normal voiding = ureterovaginal fistula (the other ureter fills the bladder)
Faeces or flatus per vaginam? Ask directly	RVF , or combined VVF/RVF; changes the operation (bowel prep, possible colostomy)
Periods; blood in the urine with periods	Cyclical haematuria + amenorrhoea + continence = vesicouterine (Youssef); amenorrhoea may also be stress, malnutrition or Sheehan's
Weak leg, dragging the foot	Foot drop : part of the injury complex; physiotherapy before surgery
Previous repair: when, where, how many	Scarring, lower success with each attempt: specialist centre
Is she pregnant or breastfeeding? Contraceptive wishes	Pregnancy test before surgery; plan contraception so she does not conceive before healing
Fluid intake	Many women drink less to leak less: dehydration, concentrated urine, UTI, stones
Social : husband, family, income, where she sleeps, transport	Reintegration and transport support; who will stay with her during the catheter period
HIV status, TB, chronic illness, nutrition	Healing and fitness for anaesthesia

Step 2: Focused examination

General: nutrition (weight, MUAC), **pallor**, hydration, smell of urine, mood, and **gait** (look for **foot drop**: a high-stepping gait, weak ankle dorsiflexion).

Abdomen: CS scar, **palpable bladder**, suprapubic tenderness, **renal angle tenderness** or a flank mass.

Vulva and perineum: **ammoniacal dermatitis**, pooling urine, perineal body present or absent, any scar from a tear or episiotomy, a gaping anus.

Speculum and digital examination (left lateral with a **Sims speculum**, or lithotomy; good light, a head torch; a chaparone):

Document	How	Why
Site	Juxtaurethral, mid-vaginal, juxtacervical, vault, cervical	Predicts route and difficulty
Distance of the distal edge from the external urethral meatus (EUM)	Measure in cm with a marked sound or finger	Goh type 1 (>3.5 cm) to type 4 (<1.5 cm)
Size	Largest diameter in cm (or fingertips)	Goh a (<1.5 cm), b (1.5–3 cm), c (>3 cm)
Scarring and vaginal length/capacity	Rigid, fixed edges; vagina >6 cm or shortened	Goh i–iii; need for a Martius flap
Urethra	Pass a metal catheter or sound up the urethra: is it intact and how long? Does the tip appear through the fistula?	Urethral involvement predicts residual stress incontinence
Edges	Slough, granulation, infection, or clean epithelialised edges	Fresh + small = catheter trial; clean = ready for repair

Document	How	Why
Cervix	Present, torn, ulcerated or a mass	A friable hard mass = biopsy (cancer)
Ureteric orifices / bladder mucosa	Seen through a large fistula; prolapsing bladder mucosa	Near the trigone = ureteric catheters at repair

Rectal examination (never skip): an **RVF** (site, size, distance from the anal verge), **anal sphincter** tone and continuity, the perineal body.

Neurological: ankle dorsiflexion power, sensation over the dorsum of the foot (common peroneal) and lateral leg.

Step 3: Bedside tests

Test	How	What it means
Palpate / catheterise the bladder	Catheter after she voids; residual volume	Residual over 150 mL = retention; dribbling may be overflow , not a fistula
Dye test	Dilute methylene blue (about 100–200 mL) into the bladder through a Foley; watch with a speculum, or place a gauze in the vagina	Blue fluid in the vagina = VVF ; note exactly where it appears
Three-swab (tampon) test	See the worked example in Step 5	Tells VVF vs ureterovaginal vs urethrovaginal
Double-dye test (where phenazopyridine is available)	Oral phenazopyridine (renal urine turns orange) + methylene blue in the bladder	Orange swab = ureteric; blue = bladder; both = combined
Cough stress test (full bladder, no fistula leak seen)	Ask her to cough while you watch the meatus	Leakage from the meatus with cough = stress incontinence
Urine dipstick	Nitrites, leucocytes, blood	UTI (treat before surgery); blood = stones, schistosomiasis or vesicouterine fistula
Pregnancy test	Urine hCG	Before surgery
HemoCue / Hb, RBS	Finger-prick	Anaemia and diabetes affect healing
HIV and syphilis RDT	If status unknown	Healing, counselling, ART
Rectal methylene blue or tampon test (suspected small RVF)	Tampon in the vagina, dilute blue (or air with the vagina under water) instilled per rectum	Blue on the tampon, or bubbles = RVF

CLINICAL PEARL

Always do a dye test before you call it a fistula. Postpartum **overflow incontinence** from a distended, atonic bladder looks exactly like a small VVF and is cured by a catheter for a few days.

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Urine microscopy and culture	Everyone before repair; symptoms of UTI	Pyuria, organism; schistosome ova in endemic areas	Treat per culture before surgery; praziquantel 40 mg/kg if ova. Culture at regional/national referral hospitals; microscopy at HC IV
FBC / Hb	Everyone before repair	Anaemia (common: blood loss, poor diet)	Iron and folate; transfuse if very low. HC IV and above
Urea, creatinine, electrolytes	Suspected ureteric injury, anuria, before repair at most centres	Raised creatinine	Suggests obstruction or bilateral injury: urgent imaging and urology. Regional/national referral
Renal and pelvic ultrasound	Suspected ureteric fistula, loin pain, recurrent UTI	Hydronephrosis, urinoma, bladder stones	Ureteric injury = urology (stent or reimplantation); stones removed at repair. Available at most HC IVs and hospitals
IVU or CT urogram	Suspected ureterovaginal fistula (wet-not-blue swab)	Contrast leak from a ureter; obstruction	Plans stenting or reimplantation. Regional/national referral (CT at national referral and some private centres)
Cystoscopy / examination under anaesthesia	Complex, high or recurrent fistulae; vesicouterine; before or at repair	Relation to the ureteric orifices and trigone; stones; extent	Surgical route and ureteric catheters. Fistula centres and regional referral hospitals
HysteroGRAPHY / cystography	Suspected vesicouterine fistula	Contrast from uterus to bladder or vice versa	Abdominal repair. Referral centres
Biopsy	Friable, ulcerated or hard mass; radiation history	Carcinoma	Changes everything: oncology, not fistula repair
Stool for ova / blood	Combined RVF, anaemia	Hookworm, schistosomiasis	Deworming (albendazole 400 mg) before surgery

Step 5: Putting it together

Model one-line summary: "A 22-year-old para 1+0 with no living child, 5 weeks after a 3-day labour ending in a CS and a fresh stillbirth, with **continuous urinary leakage since day 6, no normal voiding, right foot drop** and amniotic dermatitis; a **positive dye test** and a **2 cm mid-vaginal VVF 3 cm from the EUM with mild scarring (Goh 2bi)**: an ischaemic VVF after obstructed labour."

Problem list:

1. Ischaemic VVF, Goh 2bi (still within the 4–6-week catheter window)
2. Amniotic dermatitis
3. Right foot drop (common peroneal / lumbosacral injury)
4. Anaemia (Hb 9.2 g/dL)
5. Bereavement, low mood, rejection by her husband

Working diagnosis and plan: ischaemic VVF → Foley on free drainage, fluids 4–5 L/day, sitz baths and barrier cream, iron and folate, physiotherapy, counselling; re-examine in 1–2 weeks; refer to the fistula centre if not closed by 4–6 weeks.

Worked example 1: the three-swab test (Moir)

How to do it:

1. Empty the bladder with a Foley catheter.

- Place **three swabs (or gauze rolls) in the vagina**, one above the other: **upper** (in the vault, near the cervix), **middle**, and **lower** (just inside the introitus).
- Instil **about 100–200 mL of dilute methylene blue** into the bladder through the catheter and **clamp it** (or remove it).
- Ask her to walk around for about **10–15 minutes**.
- Remove the swabs **in order, from the lowest**, and lay them out.

Interpretation:

Finding	Meaning
Upper swab wet but NOT blue (lower swabs dry)	Ureterovaginal fistula (clear renal urine bypasses the bladder)
Upper swab blue	High VVF: juxtacervical / vault or vesicocervical
Middle swab blue	Mid-vaginal VVF
Lowest swab blue only	Urethrovaginal fistula, or dye leaking from the meatus and tracking back (a false positive): recheck with a speculum
All swabs dry after walking and coughing	No bladder fistula: think stress, urge or overflow incontinence (or a ureteric fistula that did not leak during the test; repeat if the story is convincing)
Blue and clear urine both	Combined VVF and ureteric fistula

Worked example 2: leaking after a CS, but she voids

A 30-year-old P3 had a repeat emergency CS 2 weeks ago. She now leaks continuously **but also passes normal amounts of urine**. Mild right loin ache; temperature 37.4 °C.

- Speculum:** urine pooling in the vault; no hole seen.
- Three-swab test:** upper swab **wet, not blue**; middle and lower swabs dry.
- Renal ultrasound:** **right hydronephrosis**.

Interpretation: **right ureterovaginal fistula** (surgical ureteric injury at CS). **Plan:** U&E, CT urogram or IVU where available, and **urology referral** for a ureteric stent (retrograde or antegrade) or reimplantation. A Foley catheter alone will not heal it.

EXAM FAVOURITE

"Wet but not blue" is the classic viva answer for a **ureterovaginal** fistula, and "continuous leakage with normal voiding" is the classic history.

Differential diagnosis

Condition	Distinguishing features	Key investigation
VVF	Continuous leakage; blue dye seen in the vagina	Dye test; speculum
Ureterovaginal fistula	Continuous leakage plus normal voiding ; after hysterectomy or CS; hydronephrosis	Three-swab test (swabs wet but not blue), double-dye (orange), renal US, IVU/CT urogram
Vesicouterine fistula	Cyclical haematuria, amenorrhoea, often continent; after CS	Cystoscopy, hystero-graphy, dye appearing from the os
Stress urinary incontinence	Leakage only with cough, sneeze or effort; dry at rest	Cough stress test with a full bladder; negative dye test

Condition	Distinguishing features	Key investigation
Overflow incontinence (postpartum retention)	Distended bladder, dribbling, poor stream	Palpation or bladder scan; residual over 150 mL on catheter
Overactive bladder / UTI	Urgency, frequency, dysuria	Urinalysis, culture
Vaginal discharge (vaginitis, lochia)	Not urinous, no ammonia smell	Speculum, swab, dye test negative
RVF vs anal incontinence	Faeces per vaginam (RVF) vs per anus without control (sphincter tear)	Speculum, rectal exam, methylene blue enema or tampon test

Investigations

The fistula-specific tests are summarised here. Blood tests, urine culture, imaging, cystoscopy and biopsy (with Ugandan availability) are in **Step 4 of the Clinical workup** above.

Test	What you are looking for	Why
Dye test (dilute methylene blue, about 100–200 mL, instilled into the bladder via a catheter, with a swab or gauze in the vagina)	Blue in the vagina = VVF. Note where it leaks from	Confirms and localises the fistula; separates it from other leakage
Three-swab (tampon) test	Upper swab wet but not blue = ureterovaginal; blue = VVF	Detects ureteric fistula
Double-dye test (oral phenazopyridine turns renal urine orange; methylene blue in the bladder)	Orange = ureteric; blue = bladder; both = combined	Where available

Management

A. Prevention (the most important management)

1. Primary:

- delay the first pregnancy and end child marriage
- **family planning**
- **four or more ANC contacts** (the WHO recommends 8) with **birth preparedness**
- **skilled birth attendance** and facility delivery
- **maternity waiting homes** for high-risk women

2. Intrapartum:

- **partograph or LCG monitoring**
- early referral from HC II/III
- **timely CS** (the decision-to-delivery interval matters)
- **no oxytocin in obstruction**

3. **At CS:** operate with an **empty bladder** (catheter), make a proper **bladder flap**, identify the ureters in difficult cases, and inspect the bladder before closing (instil dye if in doubt). Repair any recognised injury immediately.

4. **Postpartum:** after **prolonged or obstructed labour**, leave an **indwelling catheter for 7–14 days** with high oral fluids.

B. The fresh fistula (first 4–6 weeks): conservative management

1. **Insert a Foley catheter (size 16–18 F) on continuous free drainage.** Never clamp it. Secure it to the thigh.

2. **High oral fluid intake:** aim for **about 4–5 L a day** so the urine is pale and dilute. This reduces infection, encrustation and blockage.
3. Continue drainage for **up to 4–6 weeks**. Re-examine every 1–2 weeks. **Small fistulae may close.**
4. **Local care:**
 - Remove necrotic slough gently.
 - **Sitz baths** (warm salt water) twice daily.
 - **Barrier cream** (zinc oxide or petroleum jelly) for the vulval skin.
5. **Treat infection** according to culture.
6. **Nutrition:** protein and calories. Give **iron and folate** for anaemia.
7. **Physiotherapy:** for foot drop, and ambulation.
8. **Counselling** and hope: explain that fistula is **curable**, that it is **not a curse**, and that surgery is usually free.
9. If it has not closed by 4–6 weeks, **refer to a fistula centre** for repair once the tissues are healthy.

C. Timing of repair

- **Traditional:** wait **about 3 months** after delivery, so that oedema, infection and necrotic tissue settle and the edges are healthy.
- **Early repair:** many experienced fistula surgeons repair **as soon as the tissues are clean** (no slough, no infection). This is often during the catheter period, and closure rates are excellent. The key rule is **tissue quality, not the calendar**.
- **Iatrogenic injury recognised at operation:** **repair immediately** in layers, and catheterise for about 10–14 days.
- **Iatrogenic injury found after surgery:** a trial of catheterisation. If that fails, repair when the tissues are healthy (commonly after 6–12 weeks). A **ureteric injury** needs a urologist for a stent or reimplantation.

D. Pre-operative preparation

Item	Detail
Nutrition and anaemia	Correct anaemia: ferrous sulphate plus folic acid, and transfuse if Hb is very low.
UTI	Treat according to culture.
Deworming / schistosomiasis (endemic areas)	Albendazole 400 mg single dose; praziquantel 40 mg/kg single dose if schistosomiasis is suspected.
Physiotherapy	For foot drop and contractures.
Bowel preparation (RVF or combined)	Low-residue diet or clear fluids, and an enema or bowel preparation per the surgeon.
Consent and counselling	Chance of closure, residual incontinence, catheter duration, abstinence, future CS.
Anaesthesia	Spinal for most vaginal repairs.
Antibiotic prophylaxis	A single dose at induction (e.g. ceftriaxone 1 g IV or cefazolin 2 g IV , or gentamicin per the unit's protocol). Routine prolonged antibiotics are not needed.

E. Surgical repair principles (VVF)

1. **Route:** the **vaginal route** is used for most obstetric fistulae (well over 90%). The **abdominal route** is used for vesicouterine fistulae, high vault fistulae unreachable from below, ureteric reimplantation, and some complex cases.
2. **Position and exposure:** **exaggerated lithotomy** (or prone or knee–chest for some). Good light and suction. An episiotomy (relaxing incision) if the vagina is narrow.

3. **Identify the ureters** if the fistula is near the trigone. Catheterise the ureters if needed.
4. **Circumscribe** the fistula and **dissect widely** to separate the bladder from the vagina. Mobilise until the bladder edges come together **without tension**.
5. Excise scar sparingly. Do **not** make the hole bigger unnecessarily.
6. **Close the bladder**, usually in **one layer of interrupted absorbable sutures** (e.g. 2-0 or 3-0 polyglactin), transversely or longitudinally, whichever gives less tension.
7. **Test for water-tightness** by instilling dye through the urethral catheter. Re-suture any leak.
8. In scarred or large repairs, consider a **Martius (labial fat-pad) flap** between the bladder and vagina.
9. **Close the vagina** separately, and **do not overlap** its suture line with the bladder line.
10. **Urethral** fistulae may need **urethral reconstruction** and an anti-incontinence procedure (e.g. urethral plication, fascial sling).
11. Insert a **Foley catheter** (plus a suprapubic catheter in some centres) and a **vaginal pack** for 24–48 hours.

F. Post-operative care: "the catheter is the patient"

1. Uninterrupted bladder drainage:

- **Simple fistula: 7–10 days** (WHO 2018; seven days was as effective as 14).
- **Complex or urethral repairs:** typically **14 days** or per the surgeon.

2. Nurses check **hourly** that urine is flowing. A **blocked catheter** causes a **ruptured repair**. If flow stops, **gently flush with sterile saline**; don't wait.

3. **Oral fluids of about 4–5 L a day** (or IV if vomiting). Keep a strict **input–output chart**. The urine should be clear.

4. Remove the vaginal pack at 24–48 hours. Keep the perineum clean and dry.

5. **Early ambulation**. Give analgesia. Prevent constipation (lactulose).

6. After catheter removal:

- Timed voiding every 2–3 hours at first.
- Check residual volume.
- Start **pelvic floor muscle training**.
- Observe for leakage (a dye test if in doubt).

7. Discharge advice:

- **No sexual intercourse for about 3 months** (to let the vagina and bladder heal).
- No heavy lifting.
- Complete family planning counselling and **offer contraception**.
- **Any future delivery should be by elective CS** at a hospital.
- Attend follow-up at 3 months and at 6 months.

8. **Reintegration:** counselling, family mediation, skills training and income support (e.g. TERREWODE and camp programmes).

Drug table

Drug	Dose	Route	Frequency	Notes
Ceftriaxone (surgical prophylaxis)	1 g	IV	Single dose at induction	Or cefazolin 2 g IV. Follow the unit's protocol.
Nitrofurantoin (uncomplicated UTI, per culture)	100 mg	Oral	Every 12 hours for 5 days	Avoid if eGFR is low. Adjust to the culture result.

Drug	Dose	Route	Frequency	Notes
Metronidazole (RVF / anaerobic cover)	400–500 mg	Oral (or 500 mg IV)	Every 8 hours	Per the surgeon's protocol around RVF repair.
Ferrous sulphate + folic acid	200 mg (about 60 mg elemental iron) + 400 µg–5 mg	Oral	Once or twice daily	Correct anaemia before repair.
Albendazole	400 mg	Oral	Single dose	Deworming.
Praziquantel (schistosomiasis, endemic areas)	40 mg/kg	Oral	Single dose	Treat if haematuria or eggs are found.
Lactulose	15 mL	Oral	Every 12 hours	Soft stools after RVF or perineal repair.
Paracetamol	1 g	Oral / IV	Every 6 hours (max 4 g/day)	Analgesia. Add an NSAID if renal function is normal.
Zinc oxide cream	Thin layer	Topical	After each wash	Ammoniacal dermatitis.
Oxybutynin (urgency after repair, if persistent)	2.5–5 mg	Oral	2–3 times daily	Anticholinergic side effects; specialist advice.

G. Rectovaginal fistula

- Low RVF** (often after a missed or broken-down **third- or fourth-degree tear**):
 - a **perineal repair** (convert to a fourth-degree tear and repair in layers, including the **sphincter**), or a transvaginal layered repair
 - done when the tissues are clean, usually **about 3 months** after injury
- High or large RVF, combined VVF/RVF, or radiation-related RVF**: consider a **diverting colostomy** first, then repair, then close the colostomy.
- Around surgery**: bowel preparation, a **low-residue diet**, stool softeners (**lactulose**), and **metronidazole** cover per protocol.
- After surgery**: avoid constipation, straining and enemas, and follow up for continence.

H. When closure fails or incontinence persists

Problem	Approach
Failed closure	Re-examine at 3 months. Re-repair at a specialist centre (success falls with each attempt).
Residual stress incontinence (common with urethral involvement)	Pelvic floor muscle training; urethral plication , pubococcygeal sling or fascial sling at specialist centres.
Irreparable fistula (e.g. total urethral loss, tiny fibrotic bladder)	Urinary diversion (e.g. ureterosigmoidostomy / Mainz II pouch) in selected women after careful counselling.
Vaginal stenosis	Dilators, flaps.
Amenorrhoea / infertility	Evaluate hypothalamic causes and Sheehan's syndrome; offer fertility counselling.

What to do where

Level	Actions
HC II / III	Recognise prolonged labour and refer early . Catheterise after any long labour. Identify women leaking urine in the community, reassure them, and link to a fistula centre or camp .
HC IV / general hospital	Timely CS with good surgical technique. Catheterise for 7–14 days after obstruction. Manage fresh fistulae with a catheter, local care and nutrition. Refer non-healing fistulae. Trained surgeons may repair simple fistulae.
Regional / national referral hospital, fistula centre	Complex repairs, urethral reconstruction, ureteric injuries (urology), RVF with colostomy, diversions, training, and reintegration programmes.

Complications

- **Of the fistula itself:**
 - dermatitis
 - recurrent UTI
 - **bladder stones**
 - vaginal stenosis and dyspareunia
 - amenorrhoea and infertility
 - foot drop
 - **depression**, suicide risk, divorce and isolation
 - malnutrition (women drink less to leak less)
- **Of surgery:**
 - failed closure (about 10–20%)
 - **residual stress incontinence**
 - haemorrhage
 - **ureteric ligation**
 - UTI
 - **catheter blockage**
 - vaginal stenosis
 - urgency or overactive bladder
 - VTE
- **In future pregnancy:**
 - breakdown of the repair if she labours
 - so she needs **elective CS**
 - she also has a higher risk of obstructed labour recurring (the same pelvis)

Ward-round & exam pearls

Model presentation

"Mrs A is a 22-year-old para 1+0 from Kasese, with no living children, who presents 5 weeks after delivering a fresh stillbirth by emergency CS at a general hospital after 3 days of labour. Since day 6 postpartum she has had **continuous leakage of urine day and night** and does not void normally. She denies faecal leakage. She has **right foot drop** and has been sent away by her husband. On examination she is pale, with

ammoniacal dermatitis. Sims speculum in the left lateral position shows a **2 cm mid-vaginal VVF, with its distal edge 3 cm from the EUM, a clean edge and mild scarring (Goh 2bi)**. The dye test is positive. Rectal exam is normal. **Summary: an ischaemic VVF after prolonged obstructed labour, with foot drop and social isolation.** Plan: Foley catheter on free drainage with fluids of 4–5 L a day, perineal care, a haematinic, physiotherapy, counselling, and referral to the fistula centre for repair once the tissues are clean."

Viva questions

1. **Why does leakage start days after delivery in an ischaemic fistula?** The necrotic tissue takes 3–10 days to slough.
2. **How do you tell a ureterovaginal fistula from a VVF?** Normal voiding plus continuous leakage. On the three-swab test the swabs are wet but not blue.
3. **What is Youssef syndrome?** A vesicouterine fistula: cyclical haematuria (menouria), amenorrhoea, and continence.
4. **What are the principles of repair?** Exposure, wide mobilisation, tension-free watertight layered closure, non-overlapping suture lines, well-vascularised tissue (Martius), and **uninterrupted catheter drainage**.
5. **How long is the catheter after simple fistula repair?** 7–10 days (WHO 2018; 7 days was as good as 14).
6. **What advice about future pregnancy?** Elective CS at a hospital.
7. **What is the obstructed labour injury complex?** The ischaemic field injury: VVF/RVF, urethral loss, vaginal stenosis, foot drop, amenorrhoea, infertility, and social consequences.

OSCE tips

- In a counselling station, **start with empathy and dignity**. Use the words "it is not your fault" and "it can be treated". Explain the catheter, the surgery, abstinence, and future CS. **Ask about mood and support**.
- In a history station, always ask about **faecal leakage, voiding, foot weakness, periods, and social consequences**.

Mnemonics & memory aids

- **Timing tells the cause:** "Immediate = Iatrogenic, a week = Ischaemic, two weeks after hysterectomy = Ureter."
- **Repair: "TEAR-proof"**. Tension-free, Exposure, Adequate mobilisation, Real watertight test, Proper (separate, non-overlapping) layers, Rich blood supply (Martius), Outflow uninterrupted (catheter), Oral fluids 4–5 L, Follow-up and family planning.
- **Goh: "Site-Size-Scar"**: numbers 1–4 for site (4 = urethra, worst), letters a–c for size, Roman numerals i–iii for scarring.
- **"Catheter after every long labour"**: the cheapest fistula prevention in the world.

High-yield summary

Numbers & doses to know

Item	Value
Slough separates (ischaemic fistula leaks)	3–10 days after delivery
Catheter after prolonged or obstructed labour (prevention)	7–14 days , high fluids
Fresh fistula: continuous catheter drainage	Up to 4–6 weeks , Foley 16–18 F , never clamped

Item	Value
Oral fluids with a catheter (fresh fistula and after repair)	About 4–5 L/day
Dye test volume	About 100–200 mL dilute methylene blue
Goh site (distal edge to EUM)	Type 1 >3.5 cm ; 2 2.5–3.5 cm ; 3 1.5–<2.5 cm ; 4 <1.5 cm
Goh size	a <1.5 cm ; b 1.5–3 cm ; c >3 cm
Catheter after simple repair	7–10 days (WHO 2018); complex or urethral about 14 days
Traditional timing of delayed repair	About 3 months (but tissue quality, not the calendar, decides)
Closure rates	About 80–90% ; Ugandan cohort closure with continence about 81%
No intercourse after repair	About 3 months
Post-void residual suggesting retention	Over 150 mL
Surgical prophylaxis	Ceftriaxone 1 g IV (or cefazolin 2 g IV) single dose at induction
Praziquantel (schistosomiasis)	40 mg/kg orally, single dose
Future deliveries	Elective CS in hospital

Classic exam traps

- **"Leaking since delivery = VVF."** Not until a **dye test** is positive; overflow from retention and stress incontinence look similar.
- **A negative dye test excludes a fistula.** No: continuous leakage with a negative (clear) test is a **ureterovaginal** fistula until proven otherwise.
- **"Send her home and come back in 3 months."** Wrong for a fresh fistula: **catheterise now**; a small fresh fistula may close.
- **Clamping the catheter** or "bladder training" after repair. The catheter must drain **continuously**; a blocked catheter blows the repair.
- **Forgetting the rectal examination** and missing an RVF or sphincter injury.
- **Labelling a hard, friable, bleeding "fistula" as obstetric.** Biopsy: cervical cancer or radiation.
- **Closed = cured.** Many women still have **stress incontinence** (urethral involvement), foot drop, stenosis, infertility and social harm.
- **Future vaginal delivery after repair.** No: **elective CS**.

Questions seniors ask

- **How do you do and read a three-swab test?** Three vaginal swabs, methylene blue in the bladder, walk for 10–15 minutes, remove from below. Upper wet-not-blue = ureteric; middle blue = mid-vaginal VVF; upper blue = juxtacervical/vault; lowest only = urethral (or spill).
- **What do you measure on speculum examination?** Distance of the distal edge from the EUM, size, scarring and vaginal length, urethral length, edges (slough vs clean), plus a rectal exam: this gives the **Goh** grade.
- **Why does timing of onset matter?** Immediate = surgical injury; 3–10 days = ischaemic slough; 1–3 weeks after CS/hysterectomy = ureter.
- **What would make you refer to a specialist fistula centre?** A complex fistula: large, urethral (Goh 4), circumferential, scarred, previous failed repair, combined VVF/RVF, ureteric or vesicouterine.
- **What is Youssef syndrome?** Vesicouterine fistula after CS: cyclical haematuria (menouria), amenorrhoea, continence.

- **What is the most important nursing task after repair?** Keeping the catheter draining **every hour**, with 4–5 L fluids and a strict input–output chart.
- **What is the cheapest fistula prevention?** Timely CS, and a catheter for 7–14 days after every prolonged or obstructed labour.

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define obstetric fistula.	An abnormal communication between the genital tract and the urinary tract (VVF , urethrovaginal, vesicouterine, ureterovaginal) or the rectum (RVF), arising from childbirth. It is usually ischaemic after prolonged obstructed labour, or iatrogenic at CS or hysterectomy.
2	Mechanism of an ischaemic obstetric fistula?	The impacted fetal head compresses the bladder base, urethra, vagina (and rectum) against the pubis or sacrum, causing ischaemic pressure necrosis . The slough separates at 3–10 days , leaving a hole.
3	What does the timing of leakage onset tell you?	Immediately after surgery : direct surgical injury. 3–10 days postpartum : ischaemic (obstructed labour). 1–3 weeks after hysterectomy or CS : ureteric injury.
4	Continuous leakage but normal voiding: think?	Ureterovaginal fistula . One ureter drains into the vagina; the other still fills the bladder.
5	What is Youssef syndrome?	A vesicouterine fistula (usually after CS): cyclical haematuria (menouria) , amenorrhoea, and usually urinary continence .
6	How is the dye test done, and how is it interpreted?	Instil dilute methylene blue (about 100–200 mL) into the bladder via a catheter, with swabs in the vagina. Blue = VVF . Wet but not blue = ureterovaginal (three-swab test).
7	Double-dye test?	Oral phenazopyridine turns renal urine orange . Methylene blue in the bladder turns bladder urine blue . Orange on the swab = ureteric; blue = bladder; both = combined.
8	Goh classification: the site types.	Distance of the distal edge from the EUM : type 1 more than 3.5 cm , type 2 = 2.5–3.5 cm, type 3 = 1.5 to under 2.5 cm, type 4 under 1.5 cm (urethral, worst prognosis).
9	Goh size and scarring grades?	Size : a under 1.5 cm, b 1.5–3 cm, c over 3 cm. Scarring : i none or mild (vagina over 6 cm), ii moderate or severe fibrosis or a short vagina, iii special (radiation, ureteric, circumferential, previous repair).
10	Waldijk classification?	I : urethral closing mechanism not involved. II : involved (A without or B with (sub)total urethral involvement; a without or b with a circumferential defect). III : ureteric or other exceptional fistulae.
11	Immediate management of a fresh VVF?	A Foley catheter (16–18 F) on continuous free drainage for up to 4–6 weeks , oral fluids 4–5 L a day , sitz baths and barrier cream, nutrition, iron, treat UTI, physiotherapy, counselling. Small fresh fistulae may close.
12	What is the key principle on the timing of repair?	Tissue quality, not the calendar . Traditionally wait about 3 months. Many surgeons repair as soon as the tissues are clean, with no slough or infection. Repair a recognised intra-operative injury immediately.
13	Principles of VVF repair?	Good exposure , wide mobilisation , tension-free , watertight closure (dye test), separate, non-overlapping layers , a well-vascularised flap (Martius) if scarred, and uninterrupted bladder drainage afterwards.

#	Question	Answer
14	How long should the catheter stay after a simple VVF repair (WHO 2018)?	7–10 days. Short-term (7-day) catheterisation was as effective as longer drainage for a simple fistula. Complex or urethral repairs usually need about 14 days.
15	What is the single most dangerous post-operative event after VVF repair?	A blocked catheter : bladder distension disrupts the repair. Check the flow hourly, keep oral fluids high, and flush gently if it stops.
16	Discharge advice after fistula repair?	No intercourse for about 3 months , no heavy lifting, pelvic floor exercises, family planning, follow-up at 3 and 6 months, and elective CS for any future delivery.
17	Components of the obstructed labour injury complex?	VVF/RVF, urethral loss, vaginal stenosis, foot drop , pubic osteitis, amenorrhoea or infertility, dermatitis, bladder stones, depression, divorce and poverty.
18	Most important preventive measure after prolonged obstructed labour?	An indwelling catheter for 7–14 days with high oral fluids, plus timely CS so it never gets that far.
19	Management of a low RVF after a broken-down fourth-degree tear?	Repair when the tissues are clean (about 3 months). Convert it to a fourth-degree tear and repair it in layers, including the anal sphincter . Bowel preparation, stool softeners and a low-residue diet. A colostomy is for large, high or complex fistulae.
20	Name Ugandan fistula services.	About 29 fistula centres in 24 districts . Regional and national referral hospitals, Kitovu Hospital camps, TERREWODE Women's Community Hospital (Soroti), and the Fistula Foundation Treatment Network with the MoH. Repair is usually free.
21	How do you perform the three-swab (Moir) test?	Empty the bladder; place three swabs in the vagina (upper, middle, lower); instil about 100–200 mL dilute methylene blue into the bladder and clamp; she walks for 10–15 min ; remove the swabs from below upwards and inspect each.
22	Three-swab test: interpret each result.	Upper wet, not blue = ureterovaginal. Upper blue = juxtacervical/vault (vesicocervical). Middle blue = mid-vaginal VVF. Lowest only blue = urethrovaginal or dye spilling from the meatus. All dry = no bladder fistula: think stress, urge or overflow incontinence.
23	A woman leaks after delivery. Which bedside checks exclude a fistula mimic?	Palpate for a distended bladder and measure the post-void residual (over 150 mL = retention with overflow); a cough stress test (stress incontinence); dipstick (UTI); and a dye test (negative in all three mimics).
24	Red flags in the first weeks after CS or hysterectomy in a woman with leakage?	Fever, peritonism, loin pain or a flank mass, anuria : bladder or bowel injury with leakage into the peritoneum, or ureteric obstruction/urinoma. Urgent U&E, renal ultrasound , senior surgeon and urology .
25	Which imaging for a suspected ureterovaginal fistula, and what does it show?	Renal ultrasound (hydronephrosis, urinoma), then IVU or CT urogram (contrast leak, obstruction) at a regional/national referral hospital. Management is urological : stent or reimplantation; a Foley alone will not heal it.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

- 1.** A 19-year-old para 1 delivered a stillborn baby after 3 days of labour at home. On day 7 she noticed continuous leakage of urine. She presents on day 12. The speculum shows a 1 cm mid-vaginal VVF with sloughy edges. What is the MOST appropriate initial management?
- A Immediate vaginal repair today
 - B Continuous Foley catheter drainage with high oral fluid intake, local care and review
 - C Send her home and book a repair in 6 months
 - D Urinary diversion (ureterosigmoidostomy)
 - E Oral antibiotics and pelvic floor exercises only
- 2.** A 38-year-old had a total abdominal hysterectomy for fibroids. Two weeks later she reports continuous watery vaginal leakage but still passes urine normally. The methylene blue bladder test shows wet vaginal swabs that are not blue. What is the MOST likely diagnosis?
- A Vesicovaginal fistula
 - B Stress urinary incontinence
 - C Vaginal cuff infection
 - D Ureterovaginal fistula
 - E Vesicouterine fistula
- 3.** A 30-year-old who had two previous CS reports monthly blood in her urine since her last CS 8 months ago. She has had no vaginal periods and is continent of urine. What is the MOST likely diagnosis?
- A Vesicouterine fistula (Youssef syndrome)
 - B Bladder endometriosis
 - C Asherman syndrome
 - D Schistosomiasis haematobium
 - E Bladder carcinoma
- 4.** Three days after a simple VVF repair, the nurse notes no urine in the catheter bag for 2 hours. The patient has lower abdominal discomfort. What is the MOST appropriate immediate action?
- A Give IV furosemide
 - B Remove the catheter and ask her to void
 - C Check the tubing for kinks and gently flush the catheter with sterile saline to restore drainage
 - D Reassure her and review in the morning
 - E Clamp the catheter to train the bladder
- 5.** According to the WHO 2018 recommendation, what is the appropriate duration of bladder catheterisation after surgical repair of a simple obstetric urinary fistula?
- A 24–48 hours
 - B 3 days
 - C 21 days
 - D 42 days
 - E 7–10 days

- 6.** A fistula surgeon records: 'distal edge 1 cm from the external urethral meatus, 3.5 cm diameter, dense fibrosis, previous failed repair.' Which statement is MOST accurate?
- A This is Goh type 1ai, with an excellent prognosis
 - B This is a Goh type 4 fistula with a poor prognosis for continence, and needs a specialist centre
 - C This can be repaired by any medical officer at an HC IV
 - D This fistula will close with catheter drainage alone
 - E Classification has no prognostic value
- 7.** During an emergency CS for obstructed labour at an HC IV, the medical officer notices a 2 cm tear in the bladder dome after delivering the baby. What is the BEST management?
- A Close the abdomen and refer for fistula repair in 3 months
 - B Pack the pelvis and transfer immediately
 - C Repair the bladder in two layers with absorbable sutures, check it is watertight, and leave a catheter for about 10–14 days
 - D Insert a suprapubic drain only
 - E Leave it to heal and remove the catheter after 24 hours
- 8.** Which history finding BEST suggests that a woman's postpartum incontinence is overflow rather than a fistula?
- A Dribbling with a palpably distended bladder and a large residual on catheterisation
 - B Leakage onset on day 7
 - C Continuous leakage day and night
 - D Faeces per vaginam
 - E Blue dye on vaginal swabs
- 9.** A 20-year-old woman had a successful VVF repair 6 months ago. She is now 12 weeks pregnant. What is the MOST appropriate delivery plan?
- A Vaginal delivery with an early episiotomy
 - B Home delivery with a TBA
 - C Induction of labour at 37 weeks
 - D Elective caesarean section at a hospital near term
 - E Vacuum delivery to shorten the second stage
- 10.** Which is the MOST effective health-system intervention to prevent obstetric fistula in Uganda?
- A Routine episiotomy for all primigravidas
 - B Prophylactic antibiotics after every vaginal delivery
 - C Pelvic floor exercises in pregnancy
 - D Routine vaginal packing after delivery
 - E Skilled attendance with labour monitoring (partograph or LCG) and timely access to caesarean section
- 11.** A woman has constant urine leakage since a prolonged labour 4 months ago. The dye test confirms a 2 cm VVF 4 cm from the EUM with clean edges. Her Hb is 7.8 g/dL and urinalysis shows nitrites. What should you do BEFORE surgery?
- A Operate immediately to avoid further delay
 - B Correct the anaemia and treat the UTI according to culture, then schedule the repair
 - C Give prophylactic antibiotics for 6 weeks
 - D Wait another 12 months
 - E Perform urinary diversion

- 12.** A woman whose VVF was repaired 3 months ago has a closed fistula (negative dye test) but leaks when she coughs or stands up. The original fistula involved the bladder neck. What is the MOST likely explanation?
- A Recurrent VVF
 - B Ureterovaginal fistula
 - C Residual stress urinary incontinence from damage to the urethral closing mechanism
 - D Overflow incontinence
 - E Detrusor overactivity from UTI
- 13.** A 25-year-old reports passage of flatus and faecal material per vaginam 2 months after a vaginal delivery complicated by a perineal tear that was 'stitched in the ward'. The speculum shows a 0.5 cm opening just above the posterior fourchette. The anal sphincter is deficient anteriorly. What is the MOST appropriate plan?
- A Immediate repair today in the labour ward
 - B Diverting colostomy
 - C Metronidazole only; it will heal
 - D Vaginal packing
 - E Delayed layered repair (including the sphincter) when the tissues are healthy, with bowel preparation and stool softeners
- 14.** A woman with a repaired VVF asks when she may resume sexual intercourse. What is the BEST advice?
- A After about 3 months, once healing is confirmed at follow-up
 - B Immediately after catheter removal
 - C After 1 week
 - D Never again
 - E Only after her next delivery
- 15.** Which pre-operative finding would MOST strongly lead you to choose an abdominal rather than a vaginal approach for a urinary fistula repair?
- A A 1 cm mid-vaginal VVF
 - B A juxta-urethral fistula
 - C Mild vaginal scarring
 - D A vesicouterine fistula with cyclical haematuria after CS
 - E A fistula 4 cm from the EUM

Answers & explanations

1. B · Continuous Foley catheter drainage with high oral fluid intake, local care and review

A **small fresh fistula** should be managed with **continuous catheter drainage** (up to 4–6 weeks), fluids of 4–5 L a day, and perineal care. It may close spontaneously, and the tissues improve for repair. **A** Sloughy, necrotic edges do not hold sutures. **C** Sending her away without a catheter wastes the chance of spontaneous closure. **D** Diversion is only for irreparable fistulae. **E** Pelvic floor exercises do not close a fistula.

2. D · Ureterovaginal fistula

Continuous leakage with **normal voiding**, and **wet but not blue** swabs after hysterectomy, indicate a **ureterovaginal fistula**. **A** A VVF would stain the swabs blue. **B** SUI is effort-related, not continuous. **C** Cuff infection gives a purulent discharge, not urine. **E** There is no uterus after hysterectomy.

3. A · Vesicouterine fistula (Youssef syndrome)

Cyclical haematuria, amenorrhoea and continence after CS is **Youssef syndrome**. **B** Endometriosis can cause cyclical haematuria, but her vaginal periods would continue. **C** Asherman gives amenorrhoea without haematuria. **D** Schistosomiasis haematuria is not cyclical. **E** Carcinoma is unlikely in a young woman with this cyclical pattern.

4. C · Check the tubing for kinks and gently flush the catheter with sterile saline to restore drainage

A **blocked catheter** risks bladder distension and **breakdown of the repair**. Restore drainage immediately: check for kinks and gently flush (or replace per the surgeon's protocol). **A** Furosemide makes more urine against a blockage. **B** Early removal endangers the repair. **D** Delay is dangerous. **E** Clamping is contraindicated.

5. E · 7–10 days

WHO (2018) recommends **short-term catheterisation (7–10 days)** for simple fistula repair. A trial showed 7 days was as effective as 14, with less infection and cost. **A/B** are too short for bladder healing. **C/D** are unnecessarily long and add infection and cost.

6. B · This is a Goh type 4 fistula with a poor prognosis for continence, and needs a specialist centre

Under 1.5 cm from the EUM is **type 4**; over 3 cm is **c**; previous repair or dense fibrosis is **ii/iii**. Urethral involvement predicts failure and **residual stress incontinence**, so this is a **complex** fistula for a specialist. **A** Wrong grading. **C** Complex fistulae need expert surgeons. **D** A large, scarred fistula won't close with a catheter. **E** Goh's system predicts closure.

7. C · Repair the bladder in two layers with absorbable sutures, check it is watertight, and leave a catheter for about 10–14 days

A recognised intra-operative bladder injury should be **repaired immediately** in layers, tested for watertightness, and drained with a catheter (typically 10–14 days). **A/E** Leaving it guarantees a fistula or urinary peritonitis. **B** Transfer is unnecessary for a simple dome tear and adds risk. **D** A drain alone does not close the defect.

8. A · Dribbling with a palpably distended bladder and a large residual on catheterisation

Overflow incontinence (postpartum retention) gives dribbling with a **distended bladder** and a large residual. It is treated with catheter drainage. **B/C** suggest an ischaemic fistula. **D** suggests an RVF. **E** confirms a VVF.

9. D · Elective caesarean section at a hospital near term

After fistula repair, **elective CS** protects the repair from breakdown and avoids a repeat obstructed labour. **A/C/E** involve labour, which risks recurrence. **B** Home delivery is dangerous for anyone with this history.

10. E · Skilled attendance with labour monitoring (partograph or LCG) and timely access to caesarean section

Fistula results from **unrelieved obstructed labour**. Recognising prolonged labour and **timely CS** prevents it. **A** Episiotomy does not relieve obstruction. **B** Antibiotics do not prevent pressure necrosis. **C** Pelvic floor exercises help prolapse and SUL, not fistula. **D** Packing is irrelevant.

11. B · Correct the anaemia and treat the UTI according to culture, then schedule the repair

Optimise her first: treat the UTI and correct the anaemia (haematinics, or transfusion if needed), along with nutrition and counselling. Then repair. The fistula (Goh 1bi) is favourable. **A** Anaemia and infection impair healing. **C** Prolonged prophylaxis is unnecessary. **D** There is no reason to wait a year. **E** Diversion is for irreparable fistulae.

12. C · Residual stress urinary incontinence from damage to the urethral closing mechanism

When the fistula involved the **urethra or bladder neck**, the continence mechanism is damaged, causing **residual SUI** despite closure. It is treated with pelvic floor exercises and a plication or sling procedure. **A** The dye test is negative. **B** Leakage would be continuous. **D** Overflow gives dribbling with a full bladder. **E** Urgency, not effort, would be the trigger.

13. E · Delayed layered repair (including the sphincter) when the tissues are healthy, with bowel preparation and stool softeners

This is a **low RVF after an unrecognised or broken-down fourth-degree tear**. The treatment is **delayed layered repair** (convert to a fourth-degree tear, repair the rectal mucosa, **sphincter** and perineum) in theatre with good tissues, bowel preparation and softeners. **A** Repair in a ward with inflamed tissue fails. **B** A colostomy is for high, large or complex RVFs. **C/D** won't close an epithelialised fistula.

14. A · After about 3 months, once healing is confirmed at follow-up

Abstain for **about 3 months** to allow full healing of the bladder and vaginal repair. **B/C** are too early and risk breakdown. **D** is unnecessary and harmful to her marriage and wellbeing. **E** makes no sense, and a future delivery should be by CS.

15. D · A vesicouterine fistula with cyclical haematuria after CS

Vesicouterine fistulae (and high vault or ureteric fistulae) are usually repaired **abdominally**. **A/B/C/E** are all typically approached **vaginally**, the route for most obstetric fistulae.

Clinical cases

Work through each step before reading the model answer.

Case 1: The fresh fistula on the postnatal ward

A 17-year-old para 1+0 is on the postnatal ward of a regional referral hospital, day 8 after a CS for obstructed labour following 2 days of labour at home. The baby was stillborn. The catheter was removed on day 2. Today she is wet all the time and says urine 'just flows'.

1. What history do you take?

- **Onset** of leakage (day 6–8 suggests an ischaemic slough).
- Continuous vs effort-related.
- Whether she **voids** at all.
- **Faecal or flatus leakage**.
- Fever or loin pain.
- **Leg weakness** (foot drop).
- Mood and support: husband, family.
- Future pregnancy wishes.
- HIV status and nutrition.

2. How do you examine and confirm the diagnosis?

- Look for **ammoniacal dermatitis**.
- Palpate the abdomen for a distended bladder (to exclude overflow).
- **Sims speculum** in the left lateral position, with a good light: locate the fistula, size it, measure its distance from the EUM, and note slough and scarring.
- **Dye test**: methylene blue via catheter with a vaginal swab.
- **Rectal exam** for RVF.
- Check the foot for dorsiflexion.

3. You find a 1.5 cm sloughy VVF 3 cm from the EUM. What is your management now?

1. **Re-insert a Foley 16–18 F on free drainage** (never clamp).
2. Oral fluids **4–5 L a day**; chart input and output.
3. Sitz baths and zinc oxide cream.
4. Gentle removal of slough.
5. Urinalysis and culture, and treat UTI.
6. Hb, then haematinics.
7. Nutrition.
8. Physiotherapy.
9. **Counselling**: this is common, treatable, and not her fault.
10. Re-examine weekly for **up to 4–6 weeks**.

4. At 5 weeks the fistula is 1 cm with clean edges and has not closed. What next?

Refer to the fistula surgeon or regional centre (or the next camp) for repair now that the tissues are clean. Classify it (e.g. Goh 2ai). Optimise her Hb, treat any UTI, and give single-dose antibiotic prophylaxis at spinal anaesthesia. Plan a vaginal layered tension-free repair with a dye check.

5. What post-operative and discharge instructions will you give?

- Catheter for **7–10 days** (simple fistula), checked hourly for flow.
- Oral fluids of 4–5 L a day.
- Early ambulation.
- After removal: timed voiding and pelvic floor exercises.
- **No intercourse for about 3 months**; no heavy lifting.
- **Family planning**.
- **Elective CS** in any future pregnancy.
- Follow-up at 3 and 6 months.
- Reintegration support.

Take-away: A fresh fistula is an opportunity: catheterise continuously with high fluids and good local care. Many small ones close or become ideal for repair. The catheter is the patient.

Case 2: Leaking after hysterectomy

A 44-year-old had an emergency peripartum hysterectomy for uterine rupture at a general hospital. Ten days after discharge she returns with continuous clear vaginal leakage, but she says she still 'passes urine like before'. She also has right loin ache.

1. What is your leading differential, and why?

Right ureterovaginal fistula: a ureteric injury at hysterectomy (the ureter lies close to the uterine artery and cardinal ligament). **Normal voiding** means the left ureter still fills the bladder. The differential includes a **VVF** (vault) or a combined fistula. Loin ache suggests partial obstruction or hydronephrosis.

2. Which bedside test helps, and what are the possible results?

The **three-swab (tampon) test** with methylene blue in the bladder:

- **Wet but not blue:** ureteric.
- **Blue:** VVF.

A **double-dye test** (oral phenazopyridine plus bladder methylene blue) shows orange (ureteric), blue (bladder), or both.

3. What investigations follow?

- **Renal ultrasound** (hydronephrosis, urinoma).
- Urea, creatinine and electrolytes.
- Urinalysis and culture.
- **IVU or CT urography** to localise the leak and assess the other kidney.
- Cystoscopy with retrograde pyelography at a urology or referral centre.

4. How is it managed?

Refer to urology / the fistula centre.

- Attempt a **retrograde or antegrade ureteric stent (JJ)**. Small injuries may heal over 6 weeks.
- If obstructed and a stent fails: **percutaneous nephrostomy**, then a **ureteric reimplantation** (ureteroneocystostomy, with a psoas hitch or Boari flap if needed), usually by the **abdominal route**.
- Treat infection and protect renal function.

5. How could this have been prevented?

- Identify the ureters during difficult hysterectomy (palpate or visualise, especially near the uterine vessels and cardinal ligament).
- Clamp close to the uterus.
- Inspect for injury before closing.
- Senior help early.
- Where suspected, give IV methylene blue or indigo carmine, do cystoscopy, and repair immediately if found.

Take-away: Continuous leakage with normal voiding after pelvic surgery means ureterovaginal fistula until proven otherwise. The dye test is negative, the swabs are wet. It is a urology problem.

Case 3: Counselling the rejected woman

A 26-year-old woman from Teso is brought by a community health worker to a fistula outreach camp. She has leaked urine and stool for 4 years since a 3-day labour that ended in a stillbirth. Her husband left her. She walks with a high-stepping gait and has not menstruated since.

1. What components of the obstructed labour injury complex should you look for?

- VVF and **RVF** (combined).
- Urethral involvement.
- Vaginal stenosis.
- **Foot drop** (high-stepping gait).
- Amenorrhoea: hypothalamic, or Sheehan's if there was PPH.
- Dermatitis.
- Bladder stones.
- Malnutrition.
- **Depression**, suicidal thoughts, social exclusion.

2. Examination shows a 4 cm juxta-urethral VVF with dense fibrosis and a 2 cm mid-rectovaginal fistula. How do you classify it and where should she be treated?

A **complex combined fistula**. It is **Goh type 4 (or 3), c, ii** (under about 2.5 cm from the EUM, over 3 cm, dense fibrosis), plus an RVF. She needs a **specialist fistula centre** (e.g. a regional or national referral hospital, or TERREWODE / a partner centre), not a general HC IV.

3. What pre-operative preparation is needed?

- Nutrition and correction of anaemia.
- Treat UTI.
- Deworming, and praziquantel if schistosomiasis is endemic or haematuria is present.
- **Physiotherapy** for foot drop.
- **Bowel preparation**, and possibly a **diverting colostomy** for the RVF first.
- **Psychosocial counselling**.
- Consent covering realistic outcomes: closure is likely, but continence is less certain; there may be more than one operation; urinary diversion is a last resort.

4. How do you counsel her about her future?

- **Empathy:** it is not her fault or a curse.
- Treatment is free.
- Residual stress incontinence is possible, and further procedures (sling, plication) exist.
- Abstain from intercourse for about 3 months after repair.
- If she conceives, she needs **elective CS** at a hospital.
- Contraception.
- **Reintegration** programmes: skills and livelihood, peer support, and family mediation where she wishes.

Take-away: Fistula is a social catastrophe as much as a surgical one. Treat the whole woman: the fistulae, the foot drop, nutrition, mood and reintegration, and route complex cases to specialist centres.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.